Starter-Motor Risk-Prediction System Technical Validation & Management Review

A rigorous, evidence-backed validation of the frozen V1.1 starter-motor risk-ranking model — detection, alert lead time, maintenance windows and limits.

0.9321	13 / 14	0 / 20	168 d
Ranking AUROC (LOVO)	Failures caught	Battery-alert false alarms	Median alert lead

MODEL UNDER REVIEW
V1.1_SM (frozen; reproduced through V3.1)

REPORT DATE
02 July 2026

FLEET
34 trucks — 14 failed + 20 non-failed

METHOD
Nested LOVO Ridge · 4 features · n = 14 events

DOCUMENT TYPE
Technical validation report + Q&A dossier

STATUS
Final — for management review

Document Control

Field	Detail
Title	Starter-Motor Risk-Prediction System — Technical Validation & Management Review
Model under review	V1.1_SM starter-motor risk-ranking model (frozen)
Validation record	V1.1 (build) + V2, V2.1, V3, V3.1 (independent re-tests; benchmark reproduced)
Date	02 July 2026
Component / platform	Starter motor — BharatBenz 5528T heavy-duty truck, 24 V electrical system
Fleet	34 independent trucks: 14 failed (F) + 20 non-failed (NF)
Signals used	Existing 6-signal CAN telemetry only (VSI, SMA, RPM, CSP, ANR, GED) @ 5 s; no new hardware
Evaluation	Nested Leave-One-Vehicle-Out (LOVO) cross-validation; n = 14 failure events
Prepared by	ByteEdge — Predictive Maintenance
Classification	Confidential — for DICV management review

Purpose. This document provides the detection evidence, statistical validation, engineering interpretation and honest limits required to present the starter-motor risk-ranking system to DICV management with confidence. Every figure quoted is traceable to a committed V1.1_SM result file or one of its four independent re-test iterations; the data lineage is listed in Appendix B.

How to read it. Sections 1–3 cover what the system is and how it was built; Sections 4–5 prove it is validated and not over-fit; Sections 6–8 are the operational outputs (alerts, maintenance windows, service routing); Section 9 states the honest limits; Section 10 shows per-truck evidence; Section 11 is the path forward. A crisp companion brief for management is provided as a separate file.

A note on truck labels. Non-failed trucks keep their raw _*SM* labels throughout the body text, the tables and the per-truck evidence figures (e.g. VIN2_NF_SM stays VIN2_NF_SM); the +14 deck display-name convention is deliberately *not* applied. The single exception is the fleet-risk overview graph (Figure 6.1), which retains its native sequential display numbering for the non-failed trucks (VIN15_NF–VIN34_NF); its failed-truck labels (VIN1_F–VIN14_F) are identical to those used everywhere in this report.

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1 Executive Summary

0.9321	13 / 14	0 / 20	168 d
Ranking AUROC (LOVO)	Failures caught (recall 93%)	Battery-alert false alarms	Median alert lead

The starter-motor risk-prediction system **ranks** every truck by failure risk from existing crank- and battery-voltage telemetry, **alerts** when a degradation precursor persists, and **schedules** service inside an empirically validated maintenance window — converting reactive starter/battery breakdowns into planned maintenance with no new hardware.

What the validation confirms

- **Strong, honest ranking.** The headline is a *ranking* score (AUROC 0.9321), **not** an accuracy figure: it means **261 of 280** failed/healthy truck pairs are ordered correctly (14×20 pairs), on trucks the model never saw in training. 95% CI [0.811, 0.986]; permutation p = 0.005 (not chance).
- **Detection is strong.** The alert channels flag **13 of 14** failures, with a median first-fire lead of **168 days** (range 77–424 days before the recorded failure date). One structural blind spot remains (the VIN9-class truck).
- **Maintenance windows hold up.** Of the 11 failures that received a predicted service window, **9 fell inside it**; both misses were late on the safe side (the truck failed later than the window closed, never earlier).
- **We proved we tried to beat it.** Across three independent iterations, **17 pre-registered candidate improvements were tested and rejected**, and the frozen benchmark was reproduced to the 4th decimal four times. The model sits at the *data's* ceiling.

> **The honest ceiling — a data limit, not a method limit**
>
> Day-precise per-truck remaining life cannot beat a simple fleet-average clock, four of fourteen failures are electrically *silent*, and the confidence interval is wide — all consequences of having just **14 failure events**. Every one lifts with more data. Phase 2 (500 trucks) is the unlock, using this same validated method and no new modelling risk.

Recommendation

Deploy the system now on existing telematics: (1) weekly risk ranking and green/amber/red tiers, (2) the validated alert channels (persistence flag + A2 battery-cascade), and (3) tier-based maintenance windows for planned service. Communicate performance as **"93% ranking accuracy, 13 of 14 failures caught, zero battery-alert false alarms."** Scale to 500 trucks to unlock per-truck timing.

2 Fleet & Data

The starter-motor programme is built and validated on an **independent** fleet of 34 heavy-duty trucks — a completely different set of physical vehicles from the alternator programme (no VIN-level cross-analysis between the two is valid).

34	107.2 M	6	20,471
Trucks (14 F + 20 NF)	Telemetry rows @ 5 s	CAN signals	Catalogued crank events

Item	Detail
Fleet	34 independent trucks — 14 failed + 20 non-failed (suffix _SM)
Raw telemetry	107.2 M five-second CAN rows across the failed and non-failed cohorts
Signals (6)	VSI (supply/charging voltage), SMA (starter active), RPM, CSP (vehicle speed), ANR (engine torque), GED (excitation state)
Not available	Cranking current, battery SoC/SoH, alternator/starter temperature, GPS/region
Model unit	2,636 truck-weeks (window-anchored) + a 20,471-event gap-aware crank catalogue
Independence	Starter-motor VINs ≠ alternator VINs — different trucks; no cross-dataset inference

Because the useful starter-motor signature lives in *crank* and *resting* voltage behaviour, the pipeline first builds a per-truck crank-event catalogue and weekly voltage aggregates, then engineers change-based (not level-based) features anchored to each truck's own history. Seven trucks fall in an **SMA-dead** cohort (the starter-active signal never toggles); crank features are cohort-masked for them so a wiring/config artifact is never read as a fault.

3 The Model (Frozen V1.1)

The deployed model is a small, L2-regularised linear ranker (RidgeClassifier, alpha = 1.0) over **four** voltage-condition features. It has been **frozen since V1.1**; every later iteration (V2, V2.1, V3, V3.1) re-derives its benchmark and reproduces it to the fourth decimal. Deliberately tiny capacity is the point — at only 14 failure events a larger model would memorise, not generalise.

3.1 The four features — in business terms

Each feature is a physically interpretable proxy for starter/battery-circuit degradation. The table gives the plain-language meaning alongside the technical name used in the code and the model card.

What it measures (business)	Technical feature	Role in the model
Within-week voltage stability vs the truck's own baseline	**vsi_withinwk_std_ratio_30d_w**	The workhorse (coef +0.886); core signal
Resting (engine-off) voltage floor vs own baseline	**rest_vsi_p05_delta90**	Battery-floor sag — the battery-cascade signature
Weekly voltage-range trend (widening / narrowing swing)	**vsi_range_trend**	Directional suppressor; stabilises ranking
Crank voltage-dip trend (dip deepening near failure)	**dip_depth_last90_delta**	Crank-circuit load signature

The first two features carry most of the signal: the within-week voltage-stability term and the weekly-range trend form a **core pair selected in all 34 of 34** cross-validation folds. The resting-floor and crank-dip terms add the battery-cascade and crank-load context.

3.2 How the model produces a decision

- **Score.** Each feature is z-scored against the training trucks; the risk score is the regularised linear sum of (coefficient × z-score). Larger = more failure-like.
- **Calibrate.** A per-fold Platt step turns the raw score into a probability (calibration slope 0.86, Brier 0.124 — shippable, not rank-only).
- **Tier.** The calibrated probability maps to GREEN < 0.35 ≤ AMBER < 0.55 ≤ RED for weekly operations.

Weight basis — stated plainly for management

The four weights are **learned, not hand-set** — data-fitted Ridge coefficients, standardised so magnitudes are comparable. The modal four-feature subset is selected in **14 of 34** folds and the two core features in **34 of 34**; the model is never retrained for deployment, only re-read for explainability. No expert-tuned magic numbers drive the risk score.

4 How It Was Validated

4.1 Leave-one-truck-out, done strictly

Performance is measured under **nested Leave-One-Vehicle-Out** cross-validation: 34 rounds, and in each round the scored truck is **fully excluded** from everything. Feature screening, subset selection, the decision threshold and the probability calibration are **all redone inside each round** using only the other 33 trucks. This is why the number is deployment-grade rather than an in-sample flatterer — it is what the model would have said about a truck it had never seen.

4.2 What "0.9321" is — and is not

It is a **ranking** metric (AUROC), **not** a classification accuracy. With 14 failed and 20 healthy trucks there are 14×20 = **280** failed/healthy pairs. The model orders **261** of them correctly, with 0 ties:

AUROC = correctly-ordered pairs / total pairs = 261 / 280 = **0.9321**

The same model re-scored *in-sample* (its own training trucks) orders 262 of 280 pairs (0.9357). The gap — **262 vs 261, a single pair** — is the measured selection optimism: +0.0036. A model that had memorised its trucks would show a large in-sample-to-honest gap; a one-pair gap is the signature of a model that genuinely generalises.

Ranking vs accuracy — the precise distinction

Ranking (the headline): a pair is *correct* when score(failed) > score(healthy) — 261 of 280. **Classification (at an operating point):** a truck is *correct* when it lands on the right side of a chosen threshold (Section 6). The headline answers "does the model order trucks correctly?" — it is never a claim that the system is "93% accurate" on individual trucks.

4.3 The 10-week horizon is a detection-validity window, not a countdown

A separate walk-back test asks: how far before failure can the model still tell a failing truck from a healthy one? Re-scoring on data truncated to *k* weeks before the last record, ranking quality **holds at or above AUROC 0.75 through k = 10 weeks** and then collapses to chance at k = 11. This defines a **~10-week detection-validity window**: a flagged truck is typically within ~10 weeks of failure. It is **not** a per-truck countdown clock and must never be presented as "X weeks to failure."

Weeks before last record (k)	0	3	6	9	10	11	14
Walk-back ranking AUROC	0.936	0.921	0.843	0.818	0.768	0.704	0.625

Sustained validity holds to **k* = 10** weeks (last column above 0.75); the drop below 0.75 at k = 11 is the honest edge of the window.

Section 4 conclusion

0.9321 is a strictly cross-validated **ranking** score — 261 of 280 pairs ordered correctly on unseen trucks, with a one-pair selection-optimism gap and a validated 10-week detection horizon. It is the right headline provided it is always stated as *ranking* accuracy.

5 Why This Is Not Over-fitting

A strong number on 14 events invites one question above all others: is it real? Seven independent layers of evidence say yes. No single layer is decisive; together they are.

#	Evidence layer	What was done	Result
1	**Nested selection**	Screening, subset choice, threshold and calibration all redone inside every fold	Honest, deployment-grade number
2	**Tiny model capacity**	4 features, L2-regularised Ridge (alpha = 1.0) — chosen to resist memorising	Low variance by construction
3	**Permutation test**	Labels shuffled, full pipeline re-run (N = 200)	p = 0.005 — not chance
4	**Measured leak ceilings**	Observation-length and start-date proxies scored as upper bounds; every feature audited	n_weeks 0.952 / t_start 0.893; all features cleared
5	**Fixed-window control + banned feature**	L40 window-anchored recompute of every feature; artifact feature caught and removed	Bit-identical 0.9357 = 0.9357 (0.0 drop)
6	**Predecessor restated honestly**	V1's reported 0.921 re-scored under this stricter nested protocol	0.921 → 0.893 (optimism disclosed, 1 fake TP)
7	**Reproduced + adversarially tested**	Benchmark reproduced 4x (V2, V2.1, V3, V3.1); 17 candidate features pre-registered & tested	4x exact match; all 17 rejected

Layer 5 is worth a sentence: one early candidate (a dominant-frequency feature) turned out to be a disguised **1/n_weeks** proxy — it scored well only because failed trucks have shorter histories. It was caught by the fixed-window control, banned, and added to a binding feature registry so it can never re-enter. That is the machinery that makes the remaining 0.9321 trustworthy.

> **The one-line framing**
> The score survives a shuffle test, a fixed-window control, a leak-ceiling audit, four independent reproductions and 17 rejected attempts to beat it — and the model's own predecessor was *revised downward* under the same rules. This is a **data ceiling, not a method ceiling**.

6 Alert Channels & Operating Points

The system exposes several alert channels at **distinct** operating points. They answer different questions and must be reported **separately** — never blended into one recall/false-alarm number.

Operating point	Recall	False alarms	Median lead	Character
Classifier @ Youden threshold	13 / 14	5 / 20	—	Recall-greedy; ~43% cost saving vs run-to-failure*
RED tier only (P ≥ 0.55)	10 / 14	2 / 20	—	Low false-alarm burden (specificity 18/20)
H2 persistent-RED pager	10 / 14	0.19 ep/truck-yr	116 d	Sustained-signal pager (episode-rated)
A2 battery-cascade detector	4 / 5	0 / 20	66.5 d	Of the 5 battery-mode failures; battery-first routing

Recall denominators differ by design: the first three rows are over all 14 failures; the A2 detector's 4 / 5 is over the 5 battery-mode failures only (it does not target the others). *Cost saving is the modelled reduction in unplanned-failure cost at typical India heavy-duty cost ratios when a 13/14-recall inspection policy replaces run-to-failure; it is an economic estimate, not a validated field number.

6.1 Per-truck alert lead time

For each failed truck, the earliest validated alert lead relative to the recorded failure date (JCOPENDATE). “Persistence” is the tier-gated persistence flag; A1 is the crank-burst corroborator; A2 is the battery-cascade detector. The **median first-fire lead is 168 days** (range 77–424).

Truck	Tier	Archetype	Persistence (d)	A1 / A2 (d)	Earliest lead (d)
VIN5_F_SM	RED	A3 volatility	424	—	**424**
VIN13_F_SM	RED	A2 battery	301	A2 63	**301**
VIN11_F_SM	RED	A3 volatility	266	A1 179	**266**
VIN7_F_SM	RED	A3 volatility	266	—	**266**
VIN14_F_SM	RED	A1+A2 mixed	245	A2 28	**245**
VIN1_F_SM	GREEN	A1-then-silent	156	A1 232	**232**
VIN3_F_SM	GREEN	A2 battery	168	A2 91	**168**
VIN6_F_SM	RED	A2 battery	168	A2 70	**168**
VIN10_F_SM	RED	A1 solenoid	147	A1 160	**160**
VIN8_F_SM	RED	A4 silent	135	—	**135**
VIN12_F_SM	RED	A3 volatility	126	A1 128	**128**
VIN4_F_SM	GREEN	A4 silent	125	—	**125**
VIN2_F_SM	RED	A2 battery	77	—	**77**
VIN9_F_SM	GREEN	A4 silent	—	—	**MISSED**

Leads are retrospective, measured at the validated operating points above. Several GREEN-tier catches (VIN1, VIN3, VIN4) are recovered by the persistence/A1 channels rather than the RED tier — which is exactly why the alert layer sits on top of the tiering. Note that the silent-gap trucks carry **32–142 days of telemetry silence** before the recorded failure date, so their true lead is bounded by when data stopped, not by the model.

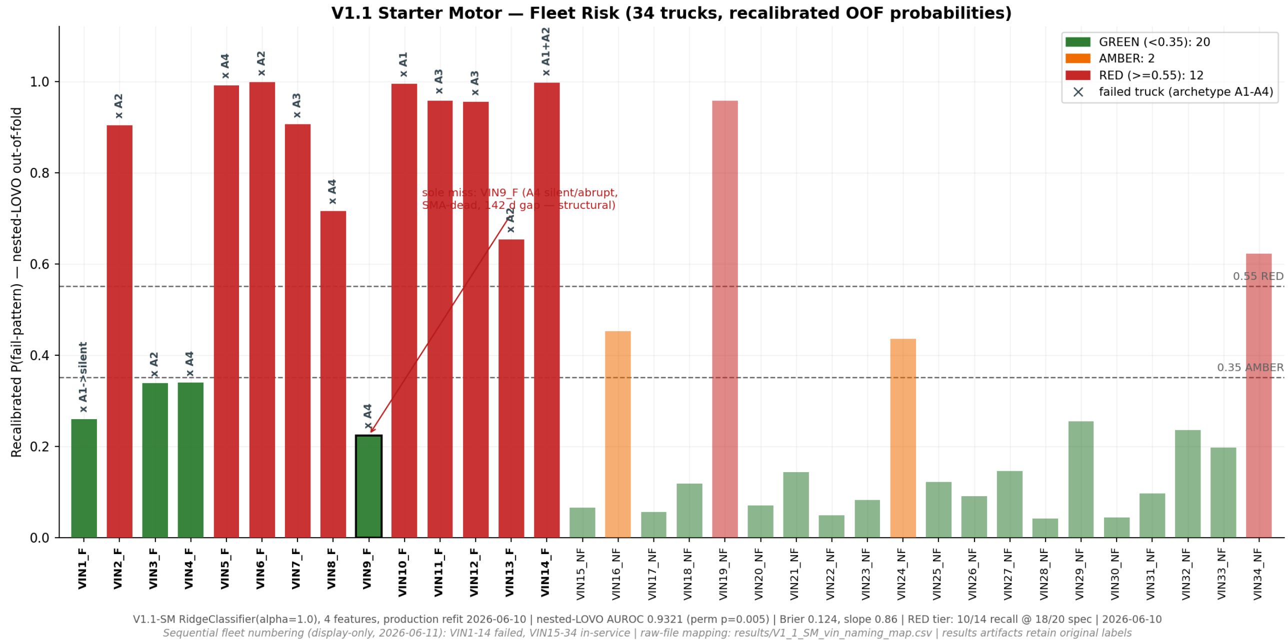

Figure 6.1 Fleet risk picture — calibrated risk across all 34 trucks; failed trucks (elevated) separate cleanly from the healthy band, with the sole structural miss (VIN9_F_SM) sitting low. Failed trucks are labelled VIN1_F–VIN14_F (identical to this report); the 20 non-failed trucks use the graph's native sequential display numbering (VIN15_NF–VIN34_NF) rather than the raw _SM labels used elsewhere here.

7 RUL as Maintenance Windows (not point estimates)

A day-precise per-truck remaining-life number would be **false precision** on this data, and we can prove it: the best per-truck survival model has an error of **576 days** against a naive constant (fleet-average) baseline of just **44 days** — the individual model is far worse than assuming every truck lasts the fleet average. So the honest, deployable deliverable is a **risk tier plus a maintenance window**, not a failure date.

7.1 Window validation

Each flagged failure is assigned a service window by band: the **battery band** (28–91 days) for A2 battery-cascade cases and the **persistence band** (126–284 days) for sustained-signal cases. A window is a *hit* when the truck's actual time-to-failure falls inside it. **9 of the 11 windowed failures landed inside**; both misses were late on the **safe side** (the truck failed after the window closed, so a scheduled service would have pre-empted it).

Truck	Tier	Band	Window (days)	Verdict
VIN2_F_SM	RED	Battery	28–91	inside
VIN3_F_SM	GREEN	Battery (A2)	28–91	late by 1 d (safe)
VIN6_F_SM	RED	Battery (A2)	28–91	inside
VIN13_F_SM	RED	Battery (A2)	28–91	inside
VIN14_F_SM	RED	Battery (A2)	28–91	inside
VIN5_F_SM	RED	Persistence	126–284	late by 140 d (safe)
VIN7_F_SM	RED	Persistence	126–284	inside
VIN8_F_SM	RED	Persistence	126–284	inside
VIN10_F_SM	RED	Persistence	126–284	inside
VIN11_F_SM	RED	Persistence	126–284	inside
VIN12_F_SM	RED	Persistence	126–284	inside

Three failures are windowless by design: **VIN1_F_SM** (GREEN, caught only by the A1 alert), **VIN4_F_SM** (AMBER — monitor, no window issued) and **VIN9_F_SM** (the structural miss). No window is ever quoted for a truck the model cannot place — that restraint is what keeps the windows honest.

Section 7 conclusion

The deployable RUL statement is **tier + maintenance window**, not a date. On this fleet the windows caught 9 of 11 with both misses on the safe side — a schedulable, auditable output that a day-precise RUL model provably cannot match at n = 14.

8 Battery-vs-Starter Service Routing (V3.1)

A recurring field question is whether a flagged truck needs the **battery** serviced or the **starter** itself. Because roughly half of starter breakdowns trace to a weak battery, the V3.1 triage (T1) routes each flagged truck to **battery-first** or **inconclusive** using the resting-voltage and battery-cascade evidence.

9 / 11	5 / 5	4 / 4	0 / 20
Triage convergence	Battery-family agreement	Silent-mode agreement	False attributions (healthy)

Of the 11 failures the triage could score (3 volatility-mode failures were left unscored by design), **9 converged** with the independently derived failure archetypes — **5 of 5** battery-family failures routed battery-first and **4 of 4** silent-mode failures flagged as inconclusive rather than mis-attributed. Critically, **0 of 20 healthy trucks** received a false battery attribution.

How to read this — SCREEN-GRADE, not diagnosis

The routing converges with *telemetry-derived* archetypes, not with confirmed workshop teardowns — so it is a **screen-grade** aid (n = 14) that tells the depot where to look first, not a warranty-grade diagnosis. Its value is the zero false attribution on healthy trucks: it never sends a good truck to battery service.

9 Known Limits (stated plainly)

The system is presented honestly. The following constraints are real; all but the last are consequences of dataset size and sensing, not of the method.

- **Fourteen failure events is the binding constraint.** The bootstrap AUROC confidence interval is [0.811, 0.986] — wide because a handful of trucks decide every threshold. This narrows only with more failures.
- **One structural blind spot (VIN9-class).** VIN9_F_SM is invisible in this telemetry: an SMA-dead (telemetry-dead) configuration, a 142-day silent gap, and an abrupt failure mode with no precursor. Its raw score (0.401) sits just under the threshold; its recalibrated probability is 0.224. This is a telemetry problem, not a modelling one.
- **Silent-gap trucks.** 5 of 14 failures carry 32–142 days of telemetry silence before the recorded failure date, which caps how much lead any model could have shown.
- **Failure dates come from workshop records** (job-card open date), so there is inherent timing uncertainty in every lead-time and window number.
- **Scores correlate with observation length** ($|r|$ up to 0.65). This is *label-mediated* — failed trucks have shorter histories because they failed — and is defended by the fixed-window (L40) control showing 0.0 AUROC borrowed from history length, plus the decay-to-chance horizon curve. The definitive cure is a larger fleet.
- **VIN independence.** Starter-motor and alternator VINs are different physical trucks; no cross-dataset, VIN-level inference is made anywhere in this report.

> **The single most important framing**
> Every limitation above lifts with more data. This is a **data ceiling, not a method ceiling**: the same validated pipeline that scores 0.9321 on 14 events will sharpen materially at 500-truck scale — and only new instrumentation (not new models) breaks the silent-failure floor.

10 Per-Truck Evidence Stacks

Every one of the 34 trucks has a per-VIN evidence stack that puts the risk-clock, the tier trajectory and the underlying voltage/crank physics on one page, so any flag can be audited truck-by-truck. Three representative examples follow; the full 34-figure set accompanies this dossier.

How to read an evidence stack

Top: the risk/RUL clock with the confirmed failure date. Middle: the calibrated tier trajectory (GREEN/AMBER/RED) with the alert channels marked where they fire. Bottom: the raw physics — resting voltage, crank-dip depth and within-week stability. **Transient elevations that self-resolve are normal; a sustained climb is the failure signature.**

10.1 VIN13_F_SM — a textbook detection

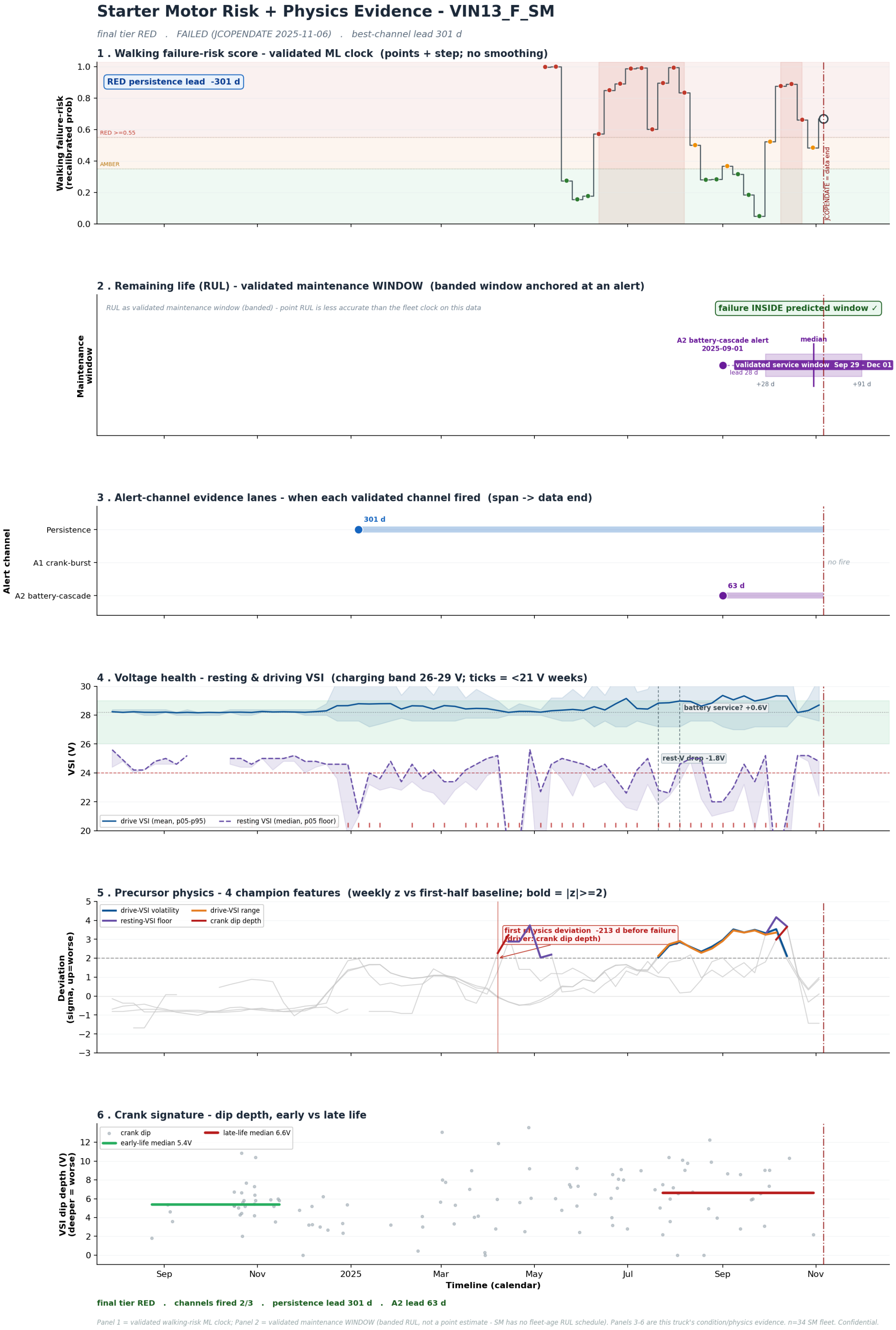

Figure 10.1 VIN13_F_SM (RED tier, battery archetype). The A2 battery-cascade detector fires 63 days before failure and the truck's time-to-failure lands inside the 28–91-day battery window — a clean, actionable, battery-first catch.

10.2 VIN9_F_SM — the honest miss

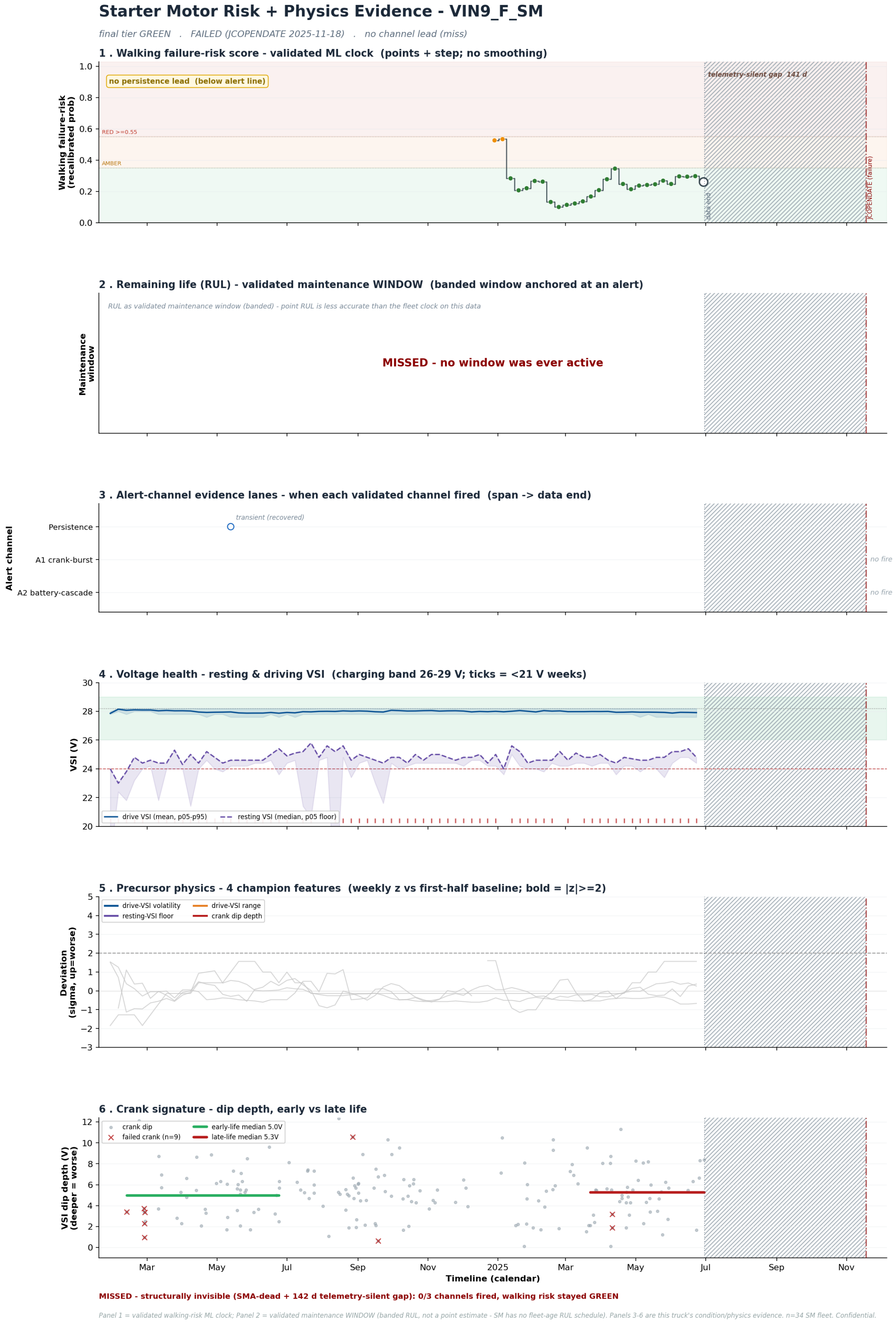

Figure 10.2 VIN9_F_SM (GREEN, MISSED). The structural blind spot: an SMA-dead configuration and a 142-day telemetry gap before an abrupt failure leave no precursor to detect. Raw score 0.401 vs threshold; recalibrated probability 0.224. We show it rather than hide it.

10.3 VIN2_NF_SM — a healthy contrast

Figure 10.3 VIN2_NF_SM (healthy). A useful contrast: the voltage signals show transient elevations that self-resolve and never build into the sustained climb that marks a failing truck — the pattern the model is trained to distinguish.

11 Path Forward

Every limit in Section 9 maps to a concrete ask. In priority order, the instrumentation and data that would break the current ceilings:

Ask	What it unlocks	Which limit it addresses
Cranking current + battery SoC / SoH sensor	Ends the battery-vs-starter ambiguity that caps alert precision	Routing is screen-grade (Sec 8)
High-frequency (≥ 1 Hz) voltage burst during cranks	Revives brush-wear / crank-dip prognosis — the biggest single unlock	5 s sampling destroys crank physics
Coolant / ambient temperature SPNs	Cold-start conditioning of the crank signature	No thermal context today
GPS / region	Environmental and duty-cycle context	No operating-environment signal
Maintenance & parts-replacement records	Turns telemetry-derived archetypes into supervised labels	Archetypes are in-sample (Sec 8)
Ignition-state signal	Cleaner crank segmentation; fewer SMA-dead ambiguities	SMA-dead cohort / VIN9-class (Sec 9)
Scale to 500 trucks (Phase 2)	~30–50+ failures — unlocks survival modelling & sharper tiers	n = 14 wide CI (Sec 9)

Bottom line for DICV management

A validated, honest, deployable system that turns reactive starter/battery breakdowns into planned maintenance — using data you already collect, with no new hardware and no methodological risk. The 0.9321 ranking score is real and conservative; the 13-of-14 detection with a 168-day median lead is actionable today; and every current limit lifts with the 500-truck scale-up using this same method.

Appendix A Question & Answer Dossier

Anticipated DICV management questions with evidence-backed answers; every number is traceable to a committed V1.1_SM result file or its re-test iterations.

Q1. Is the 0.9321 an accuracy or a ranking number?

A. It is a **ranking** metric (AUROC): draw one failed and one healthy truck at random and the model ranks the failed one higher 93.2% of the time — 261 of 280 pairs correct, measured leave-one-truck-out. It is not “correct on 93% of trucks.”

Q2. How many failures do we actually catch, and how early?

A. The alert channels catch **13 of 14** failures, with a median first-fire lead of 168 days (range 77–424). The single miss is VIN9_F_SM, a structural blind spot (telemetry-dead + 142-day gap + abrupt mode).

Q3. Are there false alarms?

A. Reported per channel, never blended. The **A2 battery-cascade detector has 0 of 20** healthy trucks flagged. The recall-greedy Youden operating point flags 5 of 20 healthy trucks; the RED tier only 2 of 20. Choose the point per maintenance economics.

Q4. Why can't you give an exact remaining-life date?

A. Because it would be false precision: the best per-truck survival model errs by 576 days versus 44 days for simply assuming the fleet average. The reliable output is a risk tier plus a maintenance window (battery 28–91 d, persistence 126–284 d).

Q5. Do the maintenance windows actually work?

A. On this fleet, 9 of the 11 windowed failures fell inside their predicted window, and both misses were late on the safe side — the truck failed after the window closed, so a scheduled service would have pre-empted it.

Q6. What is the ‘10-week horizon’ — is it a countdown?

A. No. It is a **detection-validity window**: ranking quality holds above AUROC 0.75 up to 10 weeks before the last record and then falls to chance. It means a flagged truck is typically within ~10 weeks of failure — never a per-truck ‘X weeks left’ clock.

Q7. How do we know it is not over-fit on only 14 failures?

A. Seven independent layers (Section 5): nested selection, tiny model, permutation $p = 0.005$, audited leak ceilings, a fixed-window control that reproduces the score bit-for-bit, the predecessor honestly restated 0.921→0.893, and four exact reproductions with 17 pre-registered candidate features all rejected.

Q8. Battery or starter — can the system tell which to service?

A. As a screen-grade aid, yes: the V3.1 triage routes battery-first vs inconclusive and converged with the failure archetypes on 9 of 11 scored failures (battery-family 5/5, silent 4/4) with 0 of 20 false attributions on healthy trucks. It is a where-to-look-first screen ($n = 14$), not a warranty-grade diagnosis.

Q9. What is the deployment-ready output?

A. Three things on existing telematics, no new hardware: weekly risk ranking + green/amber/red tiers, the validated alert channels (persistence flag + A2 battery-cascade), and tier-based maintenance windows for planned service.

Q10. What do you need to make it materially better?

A. In priority: cranking current / battery SoC-SoH; ≥1 Hz crank-voltage logging; maintenance & parts records (to create supervised labels); and 500-truck scale for ~30–50+ failures. Same method, more evidence.

Appendix B Data Lineage

Every figure in this report is traceable to a committed starter-motor artifact (paths relative to STARTER MOTOR/). Primary sources:

Topic	Source file
Headline metrics / CI / permutation	V1.1/results/V1_1_SM_model_spec.json · V1_1_SM_gates.json
Consolidated results	V1.1/reports/V1_1_SM_RESULTS_MASTER.md
Per-VIN nested predictions & tiers	V1.1/results/V1_1_SM_nested_lovo_predictions.csv
Alert policy & channels	V1.1/results/V1_1_SM_alert_policy.csv
10-week horizon curve	V1.1/results/V1_1_SM_horizon_curve.csv
Maintenance windows / routing (V3.1)	V3.1/heuristics/out/T2_windows.csv · T1_convergence.json · T1_attribution.csv
Fixed-window control & gates	V1.1/results/V1_1_SM_gates.json (G1–G6)
Fleet risk figure	V1.1/graphs/V1_1_SM_fleet_risk.png
Per-truck evidence stacks (34)	V1.1/visualizations/rul_evidence_stack/sm_*_evidence_stack.png
Model card / banned features	V1.1/reports/V1_1_SM_model_card.md

Prepared by ByteEdge for DICV management review. Model under review: V1.1_SM (frozen). Independent re-tests: V2, V2.1, V3, V3.1. Report generated 02 July 2026.